UNIT: I ACOUSTICS AND NDT

Introduction of Ultrasonic waves. Production of Ultrasonic waves (Piezo electric &Magnetostriction methods)-Detection (Acoustic grating), NDT Application-pulse echo method, Liquid penetrant method.

Acoustic- factor affecting Acoustic of building (Reverberation, loudness foucusing, echo, echelon effect and resonance) and their remedies- Sabine's formula for Reverberation Time-Doppler effect and its application to radars (elementary ideas)

Unit :1 ACOUSTICS & NDT

Introduction

1.1. ULTRASONIC WAVES

Sound is the mechanical vibration of the particles of material medium. A normal human ear can hear the sound waves of frequencies ranging from 20 Hz to 20,000 Hz. This frequency range is called as **audible range**.

Sound waves of frequency above the audible range (above 20,000 Hz or 20 KHz) are called **ultrasonic waves or simply ultrasonics**. These sound waves are not audible to human ear.

Ultrasonic waves are also known as high-frequency sound waves or ultrasound. Although, human ear cannot hear ultrasonic waves, surprisingly bats and some animals are capable of hearing this sound.

Bats are almost blind but can locate objects and navigate obstacles by producing ultrasonic waves (by their inborn trait) and receiving the echo. If the velocity of sound in air is 330 metre/second, the wavelength of sound waves corresponding to a frequency of 20,000 Hz is 330/20,000 = 0.0165m or 1.65 cm. So, ultrasonic waves have wavelength less than 1.65 cm in air medium. Sound waves of frequencies less than 20 Hz are called **infrasonic waves**.

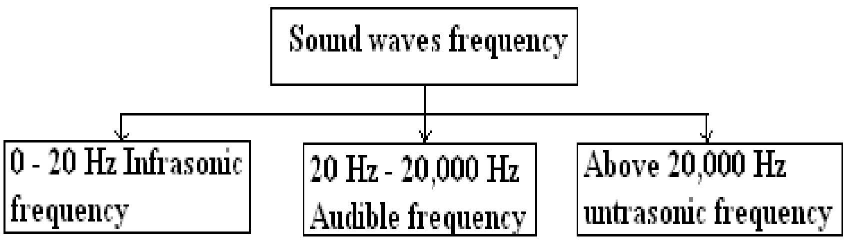

Fig.1.1. Sound waves frequency

1.2. PROPERTIES OF ULTRASONICS

a) The frequency of an ultrasonic wave is greater than 20,000 Hz.
b) It travels longer distance in the medium without any loss.
c) When an object is exposed to ultrasonics for a longer time it produces heating effect.

d) It has many modes of vibrations such as longitudinal, transverse and different modes of surface vibrations.

e) It undergoes reflection and refraction at the interface due to the change in elastic and physical properties of the medium.

1.3. PRODUCTION OF ULTRASONIC

In general there are three methods of producing the ultrasonic waves.

(i) Mechanical generator (or) Galton's whistle

(ii) Magnetostriction oscillator method

(iii) Piezoelectric oscillator method

1.3.1 MAGNETOSTRICTION GENERATOR

PRINCIPLE :

MAGNETOSTRICTION EFFECT :

The magnetostriction effect is the principle of producing ultrasonic waves.(i.e)When an alternating magnetic field is applied to a rod of ferromagnetic material such as nickel, iron, cobalt etc., alloys of it, then the rod is thrown into longitudinal vibrations thereby producing ultrasonic waves at resonance. This is known as ***magnetostriction effect***.

CONSTRUCTION:

AB is a ferromagnetic (nickel) rod clamped at the middle. L_1 and L_2 are two coils wound on the ends of the rod to set up magnetic field. The inductor coil L_1 and the variable condenser C_1 form the tank circuit.

This tank circuit is connected in between the collector and the emitter of an n-p-n transistor as shown in Fig. 1.2. Coil L_2 is connected to the base to feed back supply. The ammeter connected to see the current flow in the circuit. The input to the transistor is taken from the power supply.

WORKING:

The battery is switched ON and hence current is produced by the transistor. The current is passed through the coil L_1, which causes a corresponding change in the magnetization of the rod. Now the rod starts vibrating due to magnetostriction effect. We know that when a coil is wounded over a vibrating nickel rod, an emf is induced in the coil called **inverse magnetostriction effect**.

The induced e.m.f. is fed to the base of the transistor which acts as a feedback continuously. In this way, the current in the transistor, is built up and the vibrations of the rod is maintained. The frequency of the oscillatory circuit is adjusted by the capacitor C_1. When this frequency is equal to the natural frequency of the vibrating rod, resonance occurs.

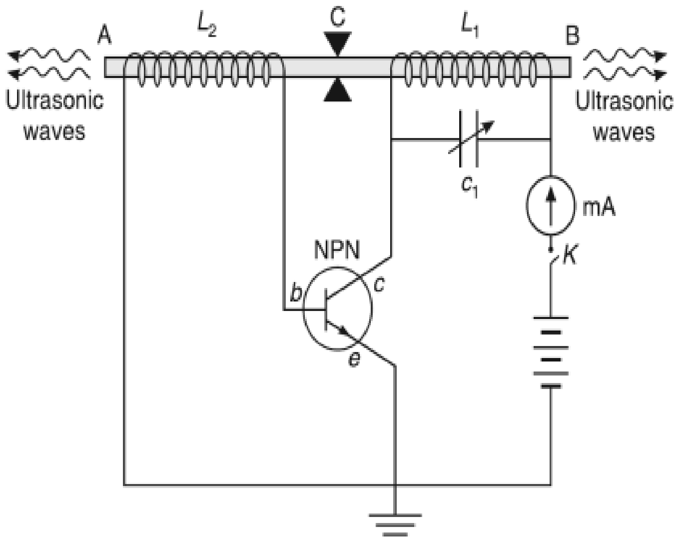

Fig.1.2. Magnetostriction generator

At resonance the rod vibrates longitudinally with larger amplitude, thereby producing ultrasonic waves of high frequency along both the ends of the rod. By adjusting the length of the rod and the capacity of the condenser for resonance, high frequencies (i.e.) oscillations of different frequency can be obtained.

Condition for resonance:

Frequency of the oscillatory circuit = Frequency of the vibrating rod

$$\frac{1}{2\pi\sqrt{L_1 C_1}} = \frac{1}{2l}\sqrt{\frac{E}{\rho}}$$

l = Length of the rod

E = Young's modulus of the rod

ρ = Density of the materials of the rod

ADVANTAGES

a) Oscillator construction is very simple and its production cost is low.

b) At low frequencies larger power output is possible without the risk of damage to the oscillatory circuit

DISADVANTAGES

a) It can produce frequencies up to 3 MHz only

b) It is not possible to get a constant single frequency because it depends on the temperature and the degree of magnetization.

c) Due to the presence of magnetic field, there will be loss of energy because of hysteresis and eddy current.

APPLICATION

Magnetostriction generator is used for high power applications such as drilling, grinding, welding etc.

1.3.2. PIEZOELECTRIC EFFECT AND PIEZOELECTRIC GENERATOR

Piezoelectric effect:When a mechanical stress is applied to one pair of opposite faces of a quartz crystal as shown in Fig.1.3. then equal and opposite electrical charges are developed on the other pair of opposite faces of the crystal. This is known as direct ***piezo-electric effect.***

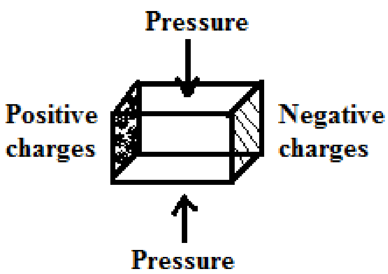

Fig.1.3.Piezoelectric effect

PRINCIPLE:

The piezoelectric generator works on the principle of inverse piezoelectric effect.

Inverse piezoelectric effect: When an electric field is applied to one pair of opposite faces of a quartz crystal, expansion or contraction (mechanical stress) is developed across the other pair of opposite faces of the crystal. This is called as **inverse piezo-electric effect** and it is utilized in the production of ultrasonic waves of high frequency.

Piezoelectric materials: The materials which exhibit Piezoelectric effect are called as Piezoelectric materials. **Example:** Quartz, Tourmalline.

CONSTRUCTION:

A thin slice of quartz crystal Q is placed in between two metal plates A and B. The plates A and B are connected to the coil L_3. The coils L_1, L_2 and L_3 are inductively coupled as shown in Fig.1.4. The coil L_1 is connected to base of the transistor circuit and coil L_2 is connected to the collector of the transistor. The capacitor C_1 is used to vary the frequency of the oscillatory circuit (L_1C_1).

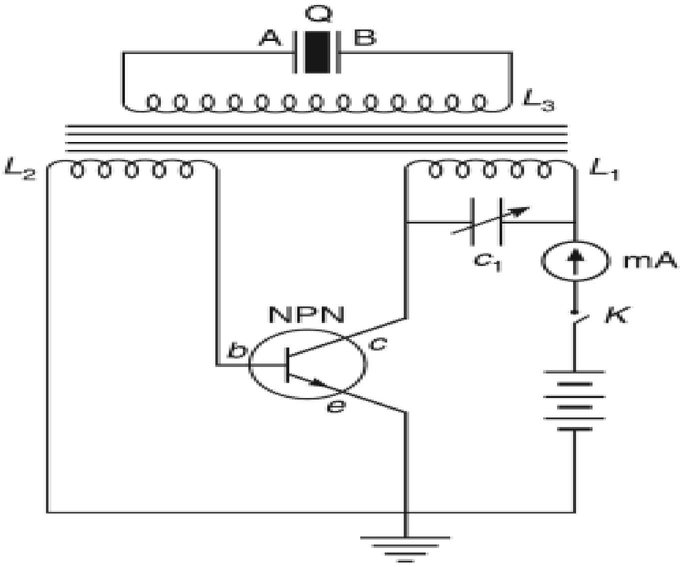

Fig.1.4. Piezoelectric generator

WORKING :

When the battery is switched ON, the current is passed through the coils L_1 and L_2 of the primary circuit. The base circuit produces alternating voltage with frequency $f = \frac{1}{2\pi\sqrt{L_1 C_1}}$.

Due to transformer action, the current is transferred to the secondary circuit and fed to the electrodes A and B. Now the crystal is under high frequency alternating voltage. Due to inverse piezo – electric effect, the crystal starts vibrating along the mechanical axis of the crystal. The frequency of the base circuit is adjusted by the variable capacitor C_1. **When this frequency is equal to the natural frequency of the vibrating crystal, resonance occurs**. At resonance the crystal vibrates longitudinally with larger amplitude, thereby producing ultrasonic waves of high frequency along both the ends of the crystal.

Condition for resonance

Frequency of the oscillatory circuit = Frequency of the vibrating crystal

$$\frac{1}{2\pi\sqrt{L_1 C_1}} = \frac{P}{2l}\sqrt{\frac{E}{\rho}} \quad P = 1,2,3 \text{(Modes of vibration)}$$

l- length of the crystal

E- Young's modulus of the crystal

ρ- density of the crystal

P =1,2,3(Modes of vibration)

ADVANTAGES

a) It is more efficient than magnetostriction oscillator.

b) It can produce frequency up to 500 MHz

c) It is not affected by temperature and humidity.

d) It produces constant frequency output.

DISADVANTAGES

a) Piezo-electric crystals are very expensive.

b) Cutting and shaping of crystals are complicated.

APPLICATIONS

It is used for low power applications such as flaw detection and medical application.

1.4. X – Cut and Y – Cut crystals:

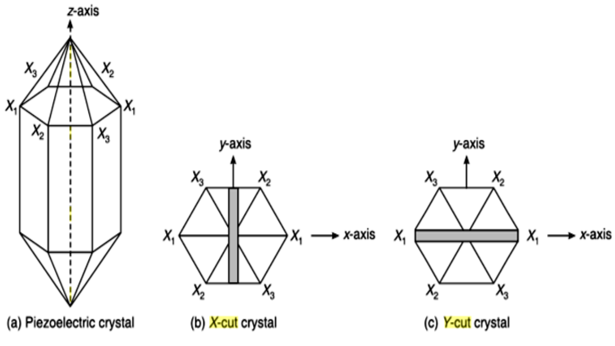

Fig.1.5. X- cut and Y – cut crystals

The piezoelectric crystals crystallize in a rhombohedron crystal structure. The rhombohedron – shaped crystal is shown in Fig 1.5.(a). The cross – sectional view of the crystal consists of six faces and looks like a hexagon and it is shown in the same Fig1.5 (b). The line joining any two opposite corners of the crystal is known as electric axis or X-axis. The line joining the midpoint of the opposite faces of the crystal is known as mechanical axis or Y-axis.

If a piece of crystal is cut perpendicular to the X-axis, then it is said to be an ***X-cut crystal***. If the crystal is cut perpendicular to the Y-axis then it is said to be ***Y-cut crystal***. The X-cut crystals are used to produce the longitudinal ultrasonic waves, whereas the transverse ultrasonic waves are produced by the Y-cut crystals.

1.5. DETECTION OF ULTRASONICS

The ultrasonic waves are detected by the following methods, namely

1. Acoustic grating method
2. Kundt's Tube Method
3. Sensitive Flame Method
4. Thermal Detectors
5. Piezoelectric detection

1.5.1. ACOUSTIC GRATING – DETERMINATION OF WAVELENGTH AND VELOCITY OF ULTRASONIC WAVES IN LIQUIDS

PRINCIPLE:

Consider a container consisting of a liquid. An ultrasonic transducer is placed in the lower end of the container. If ultrasonic waves are passed through the liquid, alternate compression and rarefactions are produced. The compressed portion acts as an opaque medium for the ordinary light and the rarefied portion acts as a transparent medium for the ordinary light. If the container is illuminated by a source of light, it undergoes diffraction and the diffraction pattern is obtained on a screen. So, the container acts as a grating. Therefore, it is said to be an acoustic grating (Fig.1.6)

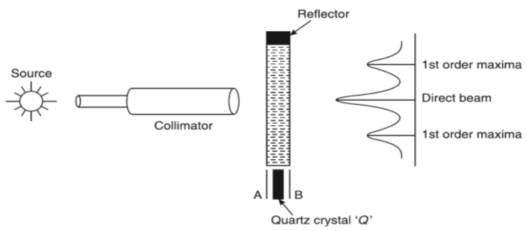

Fig.1.6. Acoustic grating

WORKING

(i) When the piezoelectric crystal is at rest:

When the quartz crystal is at rest, the light from the source is focused onto the glass tank filled with the liquid, a single image or vertical peak is observed on the screen. This shows that there is no diffraction of light.

(ii). When the piezoelectric crystal is set into vibration:

When ultrasonic waves are produced in the liquid by excitation of crystal, a number of diffracted images appear on either side of the centre. The angular separation θ between the direct image of the slit and the diffracted image of any order, say n = 1, is measured.

Calculation of ultrasonic velocity

The velocity of ultrasonic waves can be determined using the condition.

$$2d\sin\theta = n\lambda \quad \ldots(1)$$

$$\lambda = \frac{2dsin\theta}{n} \ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

w.k.t $v = f \times \lambda$

$$\lambda = \frac{v}{f} \ldots\ldots\ldots\ldots\ldots (3)$$

Substitute equ (2) in equ(1)

$$\frac{v}{f} = \frac{2dsin\theta}{n}$$

$$v = 2dsin\theta f \ldots\ldots\ldots\ldots\ldots\ldots(4)$$

Where d is the distance between any two alternate compressions or rarefactions, θ is the angle of diffraction, n is the order to diffraction and λ is the wave length of the monochromatic source of light. By knowing the frequency *f*, d and the diffraction angle θ, the wavelength and the velocity of the ultrasonic wave is calculated from Eq.(2 and 4).

1.6. ACOUSTICS

INTRODUCTION

Acoustics is defined as the science that deals with the production, control, transmission, reception, and effects of sound (as defined by Merriam-Webster). Also it is defined as the science which deals with the action of sound waves in different medium and on different obstacles. The sensation produced by stimulation of the organs of hearing by vibrations transmitted through the air or other medium is called **Sound**.

1.6.1. PROPERTIES OF SOUND

a) Sound is a form of energy.
b) Sound is produced by a vibrating body.
c) Sound requires a material medium for its propagation.
d) Sound can pass through solids, liquids and gases.
e) Velocity of sound is different in different media.
f) Sound waves undergo reflection, refraction, interference, diffraction and polarization (in transverse mode)

1.6.2. ACOUSTICS OF BUILDINGS

The branch of physics which deals with design and construction of rooms or halls, so as to give optimum acoustical efficiency is known as **architectural acoustics or acoustics of building**. The pioneering work in architectural acoustics was done by W.C. Sabine.

An auditorium or a hall is said to be acoustically good if the following conditions are satisfied.

a) The sound heard by the audience must be sufficiently loud.
b) The quality of sound should be uniform throughout the hall.
c) The syllables ought to be clear without overlapping.
d) The sounds from the exterior must not disturb the proceedings inside.
e) No echo should be present.
f) The hall should have proper reverberation time about 1.1 to 1.5 second.
g) Resonance effect (jarring effect of sound), interference or reflections should be avoided.

1.6.3. FACTORS AFFECTING THE ACOUSTICS OF BUILDINGS

The factors affecting the acoustics (sound) of buildings are as follows :

(i) Unoptimised reverberation time.

(ii) Very low (or) very high loudness.

(iii) Improper focussing of sound to a particular area, which may cause interference.

(iv) Echoes (or) Echelon effects produced inside the buildings.

(v) Resonance caused due to matching of sound waves.

(vi) Unwanted sound from outside (or) inside the building, so called noise may also affect the acoustical property of buildings.

(i) Unoptimised Reverberation time and its remedy :

We know reverberation time is the time taken for the sound to fall one millionth of as original sound intensity, when source is switched off. This reverberation time plays a vital role in the auditorium, for clear audibility of sound.

If the reverberation time is very high then it produces, echoes in the hall and if the reverberation time is very low, the sound will not be clearly heard by the audience. Therefore for clear audibility, we should maintain optimum reverberation time. The optimum reverberation time can be achieved by the following steps.

1. By having the full capacity of audience in the auditorium.
2. By choosing absorbents like felt, fibre, board, glass etc. inside the auditorium and even at the back of chairs.
3. Reverberation time can be optimised by providing windows and ventilators at the places wherever necessary and using curtains with folds for the window.
4. The reverberation time can also be optimised by decorating the walls with beautiful pictures.

(ii) Loudness

We know loundness is the degree of sensation produced on the ear, it varies from observer to observer. But, it is found that for a single observer the loudness varies from one place to another in the same auditorium. This defect is caused due to the bad acoustical construction of buildings. The loudness will be very low in some area and will be very high in some areas. It can be optimized by the following remedies.

Remedies :

a. Loudspeakers should be placed at the places where we have low loudness.

b. The loudness can also be increased by making reflecting surfaces, wherever necessary.

c. Loudness can be increased by constructing low ceilings.

d. Absorbents are placed at the places where we have high loudness.

Thus, the loudness should be made even, all over the auditorium, so that the observer can hear the sound at a constant loudness at all the places.

(iii) Focussing and Interference Effects :

In some places of a hall, the sound will not be heard properly and that place is said to be a dead space, which is due to the presence of convex or concave surfaces in the hall as shown in the below figure 1.7.

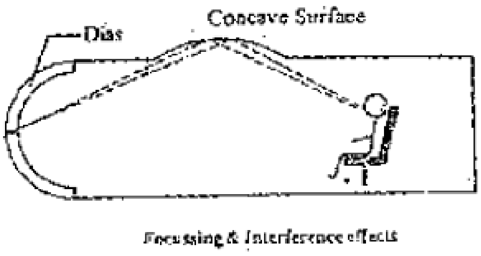

Fig.1.7.

Sometimes the sound waves will have interference pattern because of the ceiling surfaces which will create maximum intensity of sound (due to constructive interference) in some places and minimum intensity of sound (due to destructive interference) at some other places and hence causing uneven distribution of sound intensity in the hall. These problems can be rectified by the following ways.

Remedies :

1. By avoiding curved surface (or) covering the curved surfaces by suitable absorbents, the focusing can be avoided.
2. By evenly polishing and decorating with absorbents the interference effect can be avoided.

(iv) Echoes and Echelon effect :

In some halls, the walls of the halls will scatter the sound waves rather than reflecting it, this may create nuisance effect due to echoes. The echoes are formed when the time interval between the direct and reflected sound waves are about 1/15th of a second. This effect occurs due to the reason that the reflected sound waves reaches the observer later than the direct sound. If there is a regular repetition of echoes of the original sound to the observer then the effect is called as **Echelon effect** (eg.) sound produced from equally spaced steps in a stair case at regular intervals produces echelon effect.

Remedy:

The echo can be avoided by lining the surfaces with suitable sound absorbing materials and by providing enough number of doors and windows.

(v) Resonance:

Resonance occurs when a new sound note of frequency matches with standard audio frequency. Sometimes the window - panel, sections of the wooden portion are thrown into vibrations to produce new sounds, which results in interference between original sounds and create sound. This will create disturbance to the audience.

Remedies:

1. The resonance effect can be avoided by providing proper ventilation and by adjusting the reverberation time to the optimum level.

2. Nowadays the resonance is completely eliminated by air conditioning the halls.

(vi) Noise :

Noise is an unwanted sound produce due to heavy traffic outside the hall which leads to displeasing effect on the ear. There are three types of noises.

1. Air borne noise.

2. Structure borne noise.

3. Inside noise

All these three noises pollute the area at which it has been produced and create harmful effects to the human beings. Fortunately, human beings have the capability to reject the sound within certain limits with conscious efforts and to carry on his normal work. But sometimes the noises are strong which results in the following effects.

Effects produced due to noise pollution:

i. It produces mental fatigue and irritation.
ii. It diverts the concentration on work and hence reduces the efficiency of the work.
iii. Some strong noises lead to damage the eardrum and make the worker hearing impaired.
iv. The noises which are produced regularly will even retard the normal growth of infants and young children.

(i). Air borne noise :

The noise which reaches the hall through open window, doors and ventilations are called as **air borne noise**. This type of noise is produced both in rural areas [natural sound of wind and animals] and in urban areas [noise that arises from factories, aircrafts, automobiles, trains, flights etc.]

Remedies:

i. By making the hall air conditioned, this noise may be eliminated.
ii. By alloting proper places for doors and windows, this noise can be reduced.
iii. It can be further reduced by using double doors and windows with separate frames and by placing the absorbents in between them.

(ii). Structure borne noise:

The noise that reaches the hall through the structure of the building is termed as structure borne noise. These types of noises are produced inside the building. Which may be due to the machinery operation, movement of furnitures, foot steps etc., and these sound will produce structural vibration giving rise to structure borne noise.

Remedies:

i. By properly breaking the continuity of the interposing layers by some acoustical insulators this type of noise can be avoided.

ii. By providing carpets, resilient, antivibration mounts etc., this type of noise can be reduced.

(iii) Inside noise:

The noises that are produced inside the halls is known as **inside noise**. For example in some offices the sound produced by machinery, type writers etc., and produces this type of noise.

Remedies :

i. By placing the machineries and type writers over the absorbing materials.
ii. It can be reduced by covering the floors with the carpet.
iii. By fitting the engine on the floor with a layer of wood (or) felt between them this type of noise can be avoided.

1.7. REVERBERATION

When a sound pulse is produced in a hall it is reflected into a large number of times by the objects present in the hall and walls so that a series of waves of decreasing intensities fall on the listener's ears. Hence the observer does not hear a single sharp sound but a **"roll of sound"**. That is the pulse is heard for a short definite interval of time.

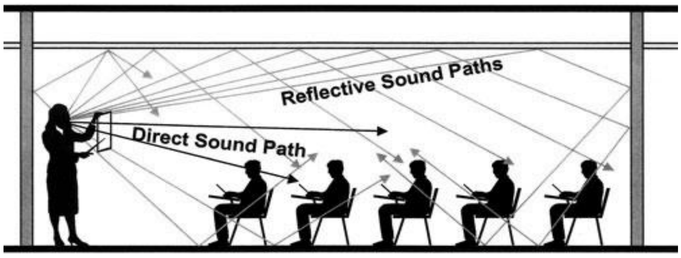

Fig.1.8. Reflection of sound inside a hall

This existence or prolongation or persistence of sound in the hall even after original source is stopped is called **Reverberation (or) lingering** of sound.

1.7.1. REVERBERATION TIME:

The duration for which sound persist is termed as **reverberation time** and is measured as the time interval between the sound produced by the source and to the sound wave until it dies.

Definition:

It is defined as the time taken for the sound to fall below the minimum audibility measured from the instant when the source stopped sounding. In designing an auditorium, theatre, conference halls ect., the reverberation time is too short it becomes inaudible by the observer and the sound is said to be dead. Therefore the reverberation time should not be too large and also it should not be too short.

So it should have an optimum value. In order to fix this optimum value a standard formula is derived by W.C. Sabine, who defined the standard reverberation time as the time taken for the sound to fall to one millionth of its original intensity just before the source is cut off. ie.,

$$E = E_m/10^6 .$$

Where E is the Energy (or) intensity of sound at any time't'. E_m is the maximum sound energy produced originally.

1.7.2. SABINE'S FORMULA FOR REVERBERATION TIME (RATE OF GROWTH AND RATE OF DECAY)

The relation connecting the reverberation time with the volume of the hall, the area and absorption co-efficient is known as **Sabine's formula**.

Sabine's developed the formula to express the rise and fall of sound intensity by the following assumptions.

(i) Distribution of sound energy is uniform throughout the hall.

(ii) There is Interference between sound waves.

(iii) The absorption co-efficient is independent of sound intensity.

(iv) The Rate of emission of sound energy from the source is constant.

Let us consider a small element 'ds' on a plane wall AB. Assume that the element 'ds' receives the sound energy 'E'. Let us draw two concentric circles of radii 'r' and 'r + dr' from the centre point 'O' of ds.

Consider a small shaded portion lying between the two semi circles drawn at an angle 'θ' and 'θ + d θ', with the normal to'ds' as shown in the Figure 1.9. Let 'dr' be the radial length and 'rdθ be the arc length. Then,

$$\text{Area of the shaded portion} = rd\theta dr \text{ ------------------------ } (1)$$

If the whole figure is rotated about the normal through an angle dϕ (Fig 1.10).

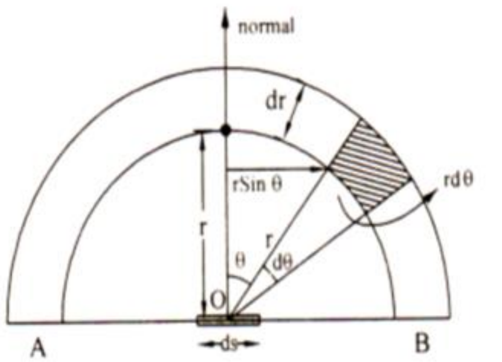

Fig 1.9

Then it is evident the area of the shaded portion travels through a small distance say 'dx'

$$dx = r\sin\theta . d\phi \text{-----------------} (2)$$

Volume traced by the shaded portion is

$$dV = \text{Area x distance}$$

Substituting from equation (1) and (2) in the above equation we have

$$dV = rd\theta dr.\ r\sin\theta.d\phi$$

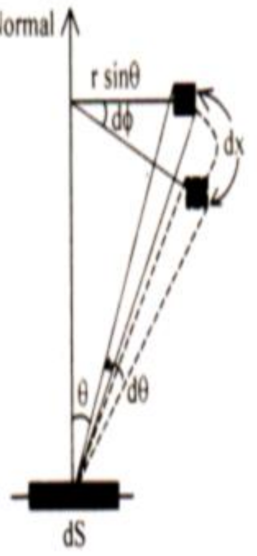

Fig.1.10.

$$dv = r^2 \sin\theta.dr.d\theta.d\phi$$

The sound energy present in this volume dv is = E X volume of the shaded portion.

$$= E\, r^2 \sin\theta.dr.d\theta.d\phi$$

This sound energy will travel through the element in all the directions.

The sound energy present in this volume 'dv' per unit solid angle is

$$= \frac{E\, r^2 sin\theta.dr.d\theta.d\phi}{4\pi}$$

Therefore in this case the soild angle subtended by the area 'ds' in the figure 1.11. at this element of volume 'dv' is $d\omega = ds'/r^2$

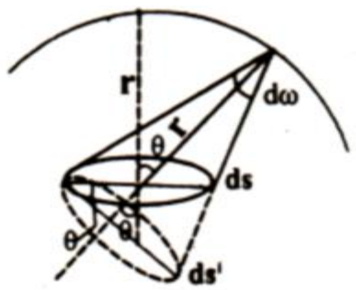

Fig1.11

$$\cos\theta = \frac{ds'}{ds} \text{ (or) } ds' = ds\cos\theta$$

Therefore we can write solid angle subtended by the area 'ds' as

$$d\omega = \frac{ds.cos\theta}{r^2}$$

Hence, the sound energy travelling from the element (i.e from dω) towards 'ds'

$$= \frac{E\, r^2 sin\theta.dr.d\theta.d\phi}{4\pi} \quad \frac{ds.cos\theta}{r^2} \qquad \ldots. (3)$$

To find the total energy received by the element 'ds' per second , we have to integrate the eq (3) for the whole volume lying within a distance 'v' of 'ds' .where v is velocity of sound.

It is obvious from the geometry of the figure that,

φ changes from 0 to 2π

θ changes from 0 to $\frac{\pi}{2}$

r changes from 0 to v

Integrating equation (3) with respect to these limits we can write energy received per second by ds is

$$ds = \frac{Eds}{4\pi}\int_0^v dr \int_0^{\frac{\pi}{2}} sin\theta cos\theta d\theta \int_0^{2\pi} d\emptyset$$

since $\int_0^{2\pi} d\emptyset = 2\pi$

$$ds = \frac{Eds\ 2\pi}{4\pi}\int_0^v dr \int_0^{\frac{\pi}{2}} sin\theta cos\theta d\theta$$

$$ds = \frac{Eds}{4}\int_0^v dr \int_0^{\frac{\pi}{2}} sin\theta cos\theta d\theta$$

$$ds = \frac{Eds}{4}\int_0^v dr \int_0^{\frac{\pi}{2}} 2sin\theta cos\theta d\theta$$

since $2sin\theta cos\theta d\theta = sin2\theta$

$$ds = \frac{Eds}{4}\int_0^v dr \int_0^{\frac{\pi}{2}} sin2\theta d\theta \ (since \int_0^{\frac{\pi}{2}} sin2\theta d\theta = 1)$$

$$= \frac{E.ds\, v}{4} \qquad ....(4)$$

Let 'a' be the absorption coefficient of the area ds.

$\therefore$ Energy absorbed by ds in unit time = $\frac{1}{4}$ E.v.a.ds

$\therefore$ Total absorption at all the surfaces of the wall= $\frac{1}{4}$ E.v.$\sum a.ds$

Total rate of energy absorption = $\frac{1}{4}$ E.v.A ....(5)

Where 'E' is the energy from sources and 'A' is the total absorption on all the surfaces on which the sound falls

(ie.) A =$\sum a.ds$

1.7.3. Growth and Decay of the sound energy

If 'P' is the power output (ie. The rate of emission of sound energy from the source) then we can write

(ie) Power output P = $\frac{1}{4}$ E_m.v.A

Here E_m is maximum energy from the source (which has been emitted) that is nothing but maximum energy which incidents on the wall

$$\therefore E_m = \frac{4P}{vA} \quad ....(6)$$

If 'V' is the volume of the hall then we can write the total energy at any instant 't' = EV

$\therefore$ Rate of growth (or) increase in energy = $\frac{d}{dt}$(EV) = V.$\frac{dE}{dt}$ ....(7)

At any instant

Rate of growth of Energy = Rate of supply of energy from the source -

Rate of absorption of energy by walls

$\therefore$ From eq (5) and eq (7) we can write

$$V.\frac{dE}{dt} = P - \frac{1}{4} EvA$$

$$\frac{dE}{dt} = \frac{P}{V} - \frac{EvA}{4V}$$

$$\text{Let } \alpha = \frac{vA}{4V}$$

$$\therefore \frac{dE}{dt} = \frac{4P}{vA}\alpha - E\alpha$$

$$(\text{or}) \ \frac{dE}{dt} + E\alpha = \frac{4P}{vA}\alpha$$

Multiplying by $e^{\alpha t}$ on both sides , we have

$$\left(\frac{dE}{dt} + E\alpha\right) \quad e^{\alpha t} = \left(\frac{4P}{vA}\right) e^{\alpha t} . \alpha$$

$$(\text{or}) \ \frac{d}{dt}(Ee^{\alpha t}) = \left(\frac{4P}{vA}\right) . \alpha . e^{\alpha t}$$

Integrating and solving we get

$$Ee^{\alpha t} = \frac{4P}{vA} . e^{\alpha t} + K \ldots\ldots.(8)$$

Where K is a constant of integration

(i) Growth of sound energy:

Let us first evaluate K for growth the boundary conditions are at t=0, E=0,

$\therefore$ eq. (8) becomes

$$0 = \frac{4P}{vA} . 1 + K (\text{or}) - \frac{4P}{vA} = K$$

Therefore substituting the value of K in eq (8) we can write

$$Ee^{\alpha t} = \frac{4P}{vA} . e^{\alpha t} - \frac{4P}{vA}$$

$$Ee^{\alpha t} = \frac{4P}{va} \quad (e^{\alpha t} - 1)$$

Since from eq. (6) $E_m = \frac{4P}{vA}$. we can write

$$Ee^{\alpha t} = E_m(e^{\alpha t} - 1)$$

Divide by $e^{\alpha t}$ throughout, we get

$$E = E_m \left(1 - \frac{1}{e^{t\alpha}}\right)$$

$$E = E_m\,(1 - e^{-\alpha t}) \qquad \ldots(9)$$

Where E_m is the maximum sound energy. This expression gives the growth of sound energy density 'E' with time't'. The growth is along an exponential curve as shown in the figure 1.12.

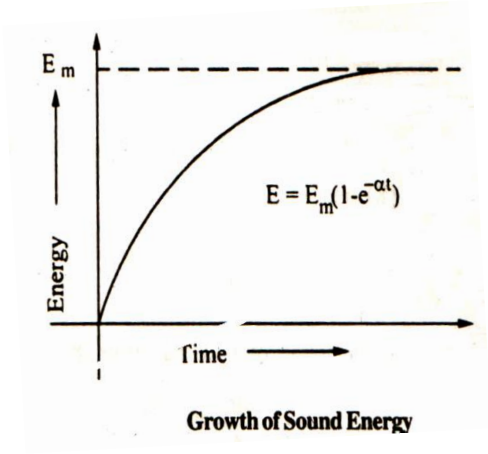

Fig.1.12.

This indicates that E increases until t = ∞ at t =∞ E=E_{max}

(ii) Decay of sound energy

Let us first evaluate K for decay. Here boundary condition are at t=0; E=E_m. Initially the sound increases from E to E_m and now it is going to decay from E_m . Therefore time is considered as zero for E=E_m(ie.). At $E = E_m$ the sound energy from the source is cut off. Therefore rate of emission of sound energy from the source = 0 (ie.) P=0.

Therefore from eq (8) we can write

$$E_m\, e^0 = 0+K \text{ (or) } K = E_m$$

Therefore substituting the value of K for decay in eq (8).we get

$$Ee^{\alpha t} = \frac{4P}{vA}.e^{\alpha t} + E_m$$

Since P=0 (ie.)Energy from source is cut off for decay of sound

We can write $Ee^{\alpha t} = E_m$

$$E = E_m e^{-\alpha t} \quad \ldots\ldots\ldots\ldots (10)$$

Eq (10) gives the decay of sound energy density with time even after the source is cut off. It is an exponentially decreasing function from maximum energy (E_m) as shown in the below figure 1.13(a) the growth and decay sound energy is also represented in the figure 1.13(b)

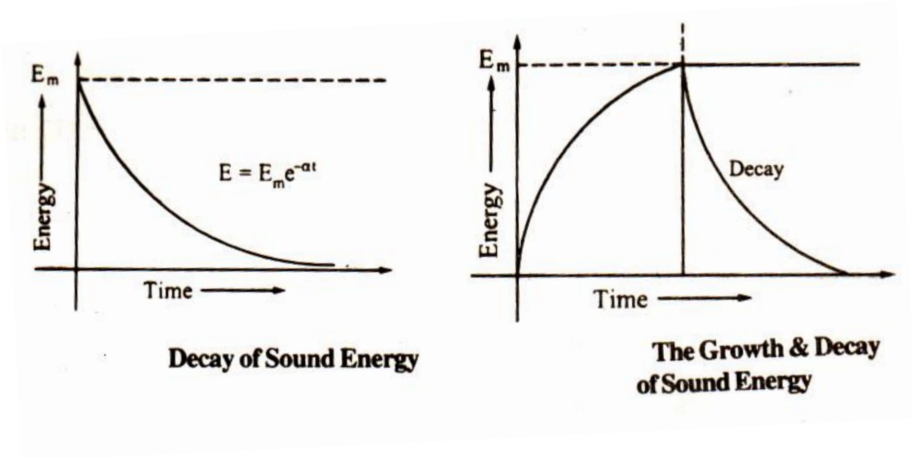

Fig.1.13 (a) **Fig1.13(b)**

1.7.4. Proof for Reverberation time (T)

According to Sabine the reverberation time is defined as the time taken by a sound to fall one millionth of its initial value when the source of sound is cut off.

The condition is , at t=T ; E $= \frac{E_m}{10^6}$

Eq (10) becomes $\frac{E_m}{10^6} = E_m e^{-\alpha t}$ (or) $10^{-6} = e^{-\alpha t}$

Take log on both sides

$$\alpha T = \log e^{10^6}$$

$$\alpha T = 6 \log e^{10}$$

$$\alpha T = 6 \times 2.3026 \log 10^{10}$$

$$\alpha T = 6 \times 2.3026 \times 1 \qquad \text{....(11)}$$

We know that

$$\alpha = \frac{vA}{4V} \qquad \text{.....(12)}$$

Substituting eq (12) in eq (11). We get

$$\frac{vA}{4V} \; T = 6 \times 2.3026$$

$$T = \frac{4V \; X \; 6X2.3026}{vA}$$

We know that velocity of sound in air at room temperature v=330m/s

$$T = \frac{4V\ X\ 6X2.3026}{330\ X\ A}$$

$$T = \frac{V\ 0.167}{A}$$

$$T = \frac{V\ 0.167}{\sum as} \qquad \ldots..(13)$$

Where $\sum as = a_1s_1 + a_2s_2 + \ldots\ldots..$

Eq (13) represents the reverberation time , which depends on three factors viz..,

(i) Volume of the hall (v)

(ii) Surface area (s)

(iii) Absorption co-efficient (a) of the materials kept inside the hall.

Among these three factors. Volume is fixed. Therefore, the reverberation time can be optimized by either varying the surface area of the reflecting surfaces or the absorption coefficient of the materials used inside the hall.

1.8. NON-DESTRUCTIVE TESTING (NDT)

As the name itself implies that NDT is a method of testing the material, without destructing (or) damaging the material, by just passing X-rays (or) ultrasonics (or) any other radiations through the material. This method is used

a) To detect the internal (or) surface flaws

b) To measure the dimensions of the materials

c) To determine the material's structure and

d) To evaluate the physical and mechanical properties of the materials.

1.8.1.ULTRASONIC FLAW DETECTOR : (PULSE ECHO METHOD)

PRINCIPLE :

Whenever there is a change in medium, the ultrasonic waves will be reflected, is the principle used in ultrasonic flaw detector. Thus, from the intensity of the reflected echoes, **the flaws are detected without destroying the material and hence this method is known as a non-destructive testing method.**

DESCRIPTION :

It consists of a Piezo electric transducer coupled to the upper surface of the specimen (metal) without any air gap between the specimen and the transducer. A frequency generator is connected to the transducer to generate high frequency pulses. The total set up is connected to the amplifier and to a cathode ray oscilloscope as shown in the figure 1.14.

i. The pulse generator generates a high potential difference and is applied to the piezo-electric transducers.
ii. The piezo-electric crystals are resonated to produce ultrasonic waves.
iii. These ultrasonic waves (pulse A) are recorded in CRO and is transmitted through the specimen.

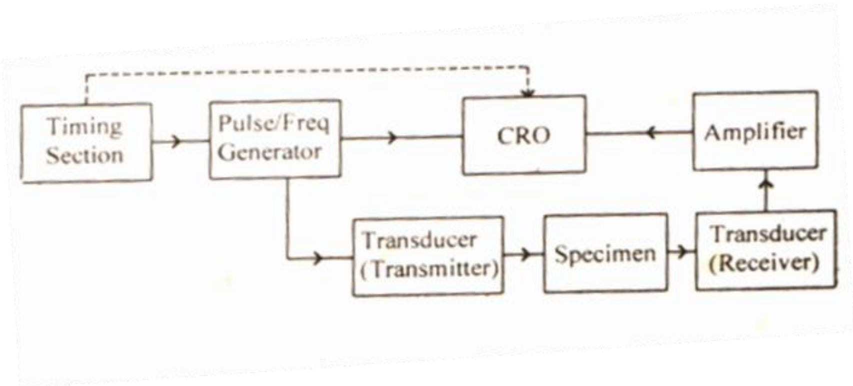

Fig.1.14

iv. These waves travel through the specimen (metal) and is reflected back by the other end.
v. The reflected ultrasonics (pulse B) is received by the transducer.
vi. These reflected signals are amplified and is found to be almost the same as that of the transmitted signals as shown in the figure 1.15 which shows that there is no defect in the specimen.
vii. If there is any defect on the specimen like a small hole (or) pores, then the ultrasonic will be reflected by the holes (ie.) defects due to the change in medium.
viii. These defects give rise to another signal (pulse Z) in between pulses 'A' and 'B'. Similarly if we have many such holes, many Z - pulses will be seen over the screen of CRO as shown in the figure 1.15.
ix. From the time delay between the transmitted and received pulses the position of the hole can be found.
x. From the height of the pulses received the depth of the hole can also be determined.

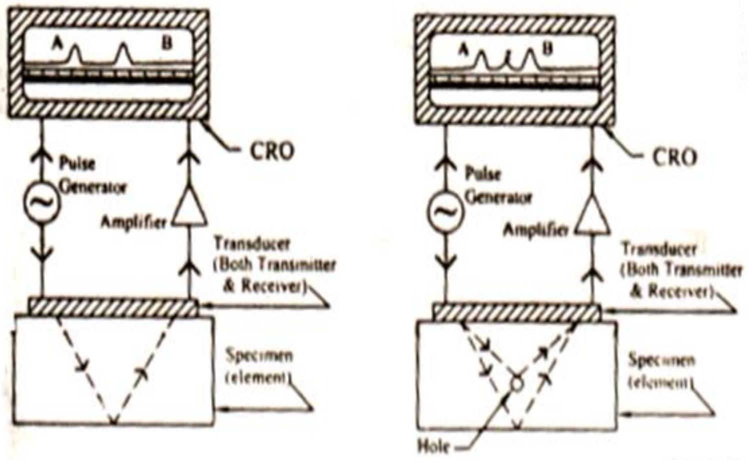

Fig. 1.15

Advantages:

1. This method is highly sensitive to most of the cracks and flaws.
2. It gives immediate results at very low cost at a very high speed.
3. It indicates the size and location of the flaws exactly.
4. Since there is no radiation in this process, it is a safest method among the other methods.

Limitations:

1. It is difficult to find the defects of the specimen which has complex shapes.
2. Trained, motivated technicians alone can perform this testing.

1.8.2 LIQUID PENETRANT METHOD

Liquid penetrant testing is the simplest non-destructive method of finding discontinuities, openings, cracks etc., in the surface of a nonporous solid material.

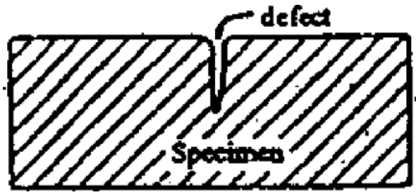

Fig1.16

Principle:

A liquid (penetrant) is applied to a cleaned surface of the specimen and sufficient time is allowed for penetration of the liquid into the defect by capillary action as shown in

Fig 1.16. After sufficient time has passed for the penetrant to enter discontinuity, the surface of the part is cleaned. The capillary action is again employed to act as a blotter to draw penetrant from the discontinuity.

Different steps involved in testing

There are five essential steps in the liquid penetrant inspection method. They are

a) Cleaning

b) Application of penetrant

c) Removal of excess penetrant

d) Application of developer

e) Inspection and evaluation

a) Cleaning

All surfaces of a specimen must be thoroughly cleaned and completely dried before it is subjected to inspection. To clean the surface, detergent and solvents like petrol or benzene etc., can be used.

b) Application of penetrant

After surface preparation, liquid penetrant is applied in a suitable manner, so as to form a film of penetrant over the component surface as shown in Fig 1.17. The liquid film should remain on the surface for a sufficient period for full penetration of the penetrant into surface defects.

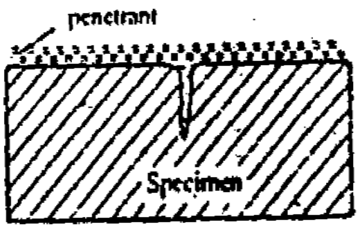

Fig.1.17

c) Removal of excess penetrant

Fig.1.18

It is now necessary to remove excess penetrant from the surface of the component. Uniform removal of excess penetrant is necessary for effective inspection but over cleaning must be avoided (Fig1.18).

d) Application of developer

The development stage is necessary to reveal clearly the presence of defects. A developer usually a very fine chalk powder is applied on the surface under inspection and it forms uniform layer over the surface of the component. The developer acts as a blotter. Penetrant liquid present within defects will be slowly drawn by capillary action into the pores of the chalk. There will be some spread of penetrant within the developer and this will magnify the apparent width of a defect. The developer also provides a uniform background to assist visual inspection (Fig.1.19)

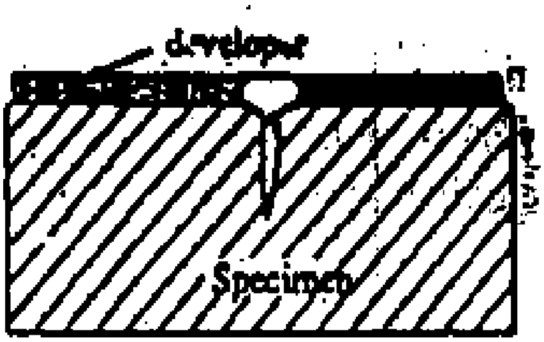

Fig. 1.19

e) Inspection and evaluation

After an optimum developing time has been allowed, the component surface is inspected for indications of penetrant 'bleedback' into the developer. Dye penetrant inspection is carried out in strong lighting conditions, while fluorescent penetrant inspection is performed in a suitable screened area using ultra-violet light. The latter technique causes the penetrant to emit visible light and defects are brilliantly outlined.

Advantages

a) The liquid penetrant process is simple
b) The equipment necessary is cheaper
c) The establishment of procedures and inspection standards for specific product parts is usually less difficult than more sophisticated methods

d) This technique can be employed for any material except porous materials

e) Penetrant inspection is suitable for components of virtually any size or shape

Limitations

a) It can detect surface-breaking defects only

b) This method is not suitable for rough surfaces and porous materials

1.9. DOPPLER EFFECT:

A change in the observed frequency of a wave, as of sound or light, occurring when the source and observer are in motion relative to each other, with the frequency increasing when the source and observer approach each other and decreasing when they move apart as shown in Fig.1.20. The motion of the source causes a real shift in frequency of the wave, while the motion of the observer produces only an apparent shift in frequency. Also called **Doppler *shift***.

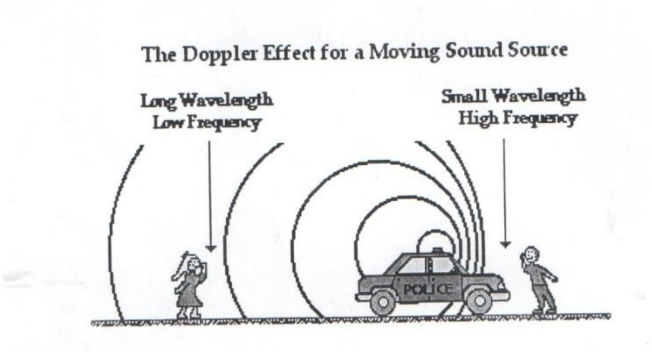

Fig. 1.20.

1.9.1. DOPPLER EFFECT AND ITS APPLICATIONS TO RADAR:

The basic concept of radar is relatively simple even though in many instances its practical implementation is not. Radar operates by radiating electromagnetic energy and detecting the echo returned from reflecting objects(targets). The nature of the echo signal provides information about the target. The range, or distance, to the target is found from the time it takes for the radiated energy to travel to the target and back. The angular location of the target is found with a directive antenna (one with a narrow beam width) to sense the angle of arrival of the echo signal. If the target is moving, radar can derive its track, or trajectory, and predict the future location. The shift in frequency of the received echo signal due to the Doppler effect caused by a moving target allows a radar to separate desired moving targets (such as aircraft) from undesired stationary targets (such as land and sea cutter) even though the stationary echo signal may be many orders of magnitude greater than the moving target.Radar is an active device in that it carries its own transmitter and does not depend on ambient radiation, as do most optical and infrared sensors.

Radar can detect relatively small targets at near or far distances and can measure their range with precision in all weather, which is its chief advantage when compared with other sensors. The principle of radar has been applied from frequencies of a few megahertz to well beyond the optical region (laser radar). Radar was originally developed to satisfy the needs of the military for surveillance and weapon control. Military applications have funded much of the development of its technology. However, radar has seen significant civil applications for the safe travel of aircraft, ships, and spacecraft; the remote sensing of the environment, especially the weather; and law enforcement and many other applications.

1.9.2. APPLICATIONS OF RADAR IN WEATHER FORECASTING:

Radars are most commonly used in detecting and predicting weather conditions. Most weather radars use the Doppler Effect while there are still a few using microwave signals. These radars are much more accurate than the older style radars. In Doppler radars the signal is sent out a constant rate. This signal is then shifted according to the Doppler Effect as it returns. The amount of the signal shift depends up on the speed of the target.

This signal is picked up and interpreted with greater accuracy than the other radar. These radars are considered to be the best tool for detection of rainfall in an area. The way they work is by sending out beams in a circular pattern around the radar. When any of these beams strikes a raindrop, part of this particular beam will be reflected back to the radar. The radar then receives the reflected beam and calculates the distance away and its intensity and from the direction it is moving and the wind velocity. The radar then translates the distance and intensity on to a graphical map that is displayed on TV.

1.9.3. APPLICATIONS OF RADAR IN LAW ENFOREMENT:

The Doppler Effect is used in some types of radar to measure the velocity of detected objects. A radar beam is fired at a moving target e.g. a motor car, as police use radar to detect speeding motorists as it approaches or recedes from the radar source. Each successive radar wave has to travel farther to reach the car, before being reflected and redetected near the source. As each wave has to move farther, the gap between each wave increases, increasing the wavelength. In some situations, the radar beam is fired at the moving car as it approaches, in which case each successive wave travels a lesser distance, decreasing the wavelength. In either situation, calculations from the Doppler Effect accurately determine the car's velocity.

Pondicherry University 2 marks Question and Answer

1. What do you mean by absorption co-efficient? (Jan 2013, Jan11)

The absorption co-efficient of a material is defined as the ratio of sound energy absorbed by it's surface to that of the total sound energy incident on the surface.

2. Suggest any two ways of reducing the time of reverberation?(Jan 2013)

(i) By providing many windows and ventilators (ii) Using heavy curtains with folds

3. What do you mean by attenuation of ultrasonic Waves? (May 2012, April/May2011)

If ultrasonic waves passed through a medium the measure of intensity per unit time per unit area perpendicular to the direction of propagation decreases with distance from source. This phenomenon is known as attenuation of ultrasonic Waves

4. Define Reverberation time (Jan11 , Jan14, Jan 16, April 10, April / May 2011, 12, 17)

The persistence of audible sound, even after the source has stopped to emit the sound is called reverberation. The time during which the sound persists in the hall is called as reverberation time.

5. What is Piezo Electric Effect? (Jan 2014, May 2012)

When pressure or mechanical force is applied along certain axis (Mechanical axis) with respect to optic axis of the crystal like quartz, tourmaline, rochelle salts etc. than equal and opposite charges are produced along the axis (electrical axis) perpendicular to the optic axis of the crystal. This effect is called Piezoelectric effect.

6. What is meant by Acoustics of Buildings? (Jan10, Jan11, May12)

Acoustics is a branch of Physics that deals with design, generation, detection, and propagation of sound waves in buildings is called acoustics of buildings.

7. What is magnetostriction effect? (May2013)

Magnetostriction effect is the principle of producing ultrasonic waves. (ie.) when an alternating magnetic field is applied to a rod of ferromagnetic material such as nickel, iron, cobalt etc., or alloys of it, then the rod is thrown into longitudinal vibrations, thereby producing ultrasonic waves at resonance.

8. What is meant by Echo? (May2013)

When a reflecting surface is far away from the source (more than 17m) the sound is reflected back as a distinct repetition of direct sound, the reflected sound is called an echo

9. What are transverse ultrasonic waves? (Jan 2010, Jan11, May 2017)

Ultrasoinc waves having vibrations perpendicular to the direction of propagation are called as transverse ultrasonic waves.

10. What is Acoustic grating? (Jan 2014, Jan15)

When ultrasonic waves are passed through liquid, the density of the liquid varies layer by layer due to the variation in pressure. This periodic variation in density results in the change of refractive index of the liquid and hence the liquid act as a diffraction grating so called acoustic grating. Under this condition when a monochromatic source of light is passed through the acoustical grating, the light gets diffracted.

11. List out any four factors affecting the acoustic quality of a building. (Jan14)

Reverberation Time, Loudness, Focusing, Echoes, Echelon Effect, Resonance, Noises

12. What are ultrasonic waves? (April 2010, May13)

Sound waves having frequencies greater than 20KHz are called transverse ultrasonic waves.

13. What is Echelon effect? (Jan 2015)

If there is a regular repetition of echoes of the original sound to the observer then the effect is called as Echelon effect

14. What is the main difference in the quality of ultrasonic waves produced by piezoelectric and magnetostriction methods? (Jan15)

Using Magneto striction method, frequencies ranging from few hundred
Hertz to about 300 KHz can be produced. Using Piezo- Electric effect,
Frequencies as high as 500 MHz can be produced.

15. Mention the name of few universally accepted common methods of NDT. (Jan10)

There are two types of methods of NDT. They are,

i) Pulse echo method (Ultrasonic flaw detector)

ii) Liquid penetrating method

16. What are X-Cut and Y-Cut crystals. (Jan13)

X-Cut Crystal: When the crystal is cut perpendicular to the X- axis then it is called X-Cut crystal.

Y-Cut Crystal: When the crystal is cut perpendicular to the Y-axis then it is called Y-Cut Crystal.

17. Mention some the uses of ultrasonic waves. (Jan2014)

a) They are used in welding and soldering
b) They are used for cleaning cloths and tiny parts of complicated machines
c) To detect flaws in materials
d) For the measurement of the depth of the sea
e) Ultrasonics are used for detecting tumors and other defects in human body
f) It is used to find the fetal heart movement and blood circulation
g) It is used to find the velocity of blood flow
h) Ultrasonics are used to remove kidney stones and brain tumors without any loss of blood.
i) Ultrasonic therapy can be used to treat disease like neuralgic and rheumatic pains etc.,

18. Mention any four characteristic features required for the selection of penetrant used in liquid penetrant method. (Jan2012).

(i) High fluidity
(ii) Lower viscosity
(iii) Higher wetability
(iv) Non- toxic
(v) Cheap

19. What is standard reverberation time? (or) What is Sabines's law? Jan 2016)

Sabines's law states that the standard reverberation time is the time taken by the sound to fall from one millionth of its original intensity, after the source of sound is stopped $I = I_0 / 10^6$

20. What is Loudness? Give the relation between loudness and intensity of sound. (May 2017)

Loudness is the degree of sensation produced on the ear. This cannot be measured directly. So that it is measured in terms of intensity. Loudness is proportional to the logarithmic value of intensity

$$L = \log I$$

$L = K \log I$ this is also known as Weber-Fechner's Law.

21. What are sound absorbing materials?(May 2017)

Materials which have the ability to absorb sound waves to provide good quality of sound are called as sound absorbing materials. Examples Carpets, Open window hair felt, ordinary chair, glass wooden chairs etc.,

22. What is the basic principle of ultrasonic pulse echo method? (May 2017)

Whenever there is a change in medium, the ultrasonic waves will be reflected, is the principle used in ultrasonic flaw detector. Thus, from the intensity of the reflected echoes, the flaws are detected without destroying the material and hence this method is known as a non-destructive testing method.

Additional 2 marks Question and Answer

Part - A

1. How are sound waves classified ?

With respect to frequency sound waves are classified into three categories they are

(a). Infrasonic – Frequency less than 20 Hz.

(b). Audible Range – Frequency between 20 Hz to 20000Hz

(c). Ultrasonic – Frequency above 20000 Hz

2. Name methods by which ultrasonic waves are produced.

In general there are three methods of producing the ultrasonic waves.

(i) Mechanical generator (or) Galton's whistle

(ii) Magnetostriction oscillator method

(iii) Piezoelectric oscillator method

3. Explain the properties of ultrasonic waves.

(a) The frequency of an ultrasonic wave is greater than 20,000 Hz.

(b) It travels longer distance in the medium without any loss.

(c) When an object is exposed to ultrasonics for a longer time it produces heating effect.

(d) It has a many modes of vibrations such as longitudinal, transverse and different modes of surface vibrations.

(e) It undergoes reflection and refraction at the interface due to the change in elastic and physical properties of the medium.

4. What is meant by optimum reverberation time? Give its value for concert hall and theatres.

A satisfactory or preferred value of the reverberation time is called **Optimum reverberation time.**

The satisfactory reverberation times are

Speeches	0.5 second
Music	1 to 2 second
Theatres	1.1 to 1.5 second

5. Why not ultrasonics be produced by passing high frequency altering current through a loud speaker?

Ultrasonics cannot be produced by passing high frequency alternating current through loudspeakers due to following reasons.

1. Loud speaker coil cannot vibrate at such high frequency.
2. Inductance of the speaker coil becomes so high and practically no current flows through it.

6. What are the methods used for the detection for Ultrasonics?

1. Kundt's tube method
2. Sensitive flame method
3. Thermal detector
4. Piezo-electric detector
5. Acoustic Grating

7. Write down Sabine's formula for reverberation time.

Reverberation time T = 0.165 V/∑as

Where V – Volume of the hall in m^3

a – Absorption coefficient of surface areas of different material in O.W.U

S- Surface area of the different materials in m^2

∑as – Total absorption of sound in O.W.U m^2

8. What is meant by resonance effect in acoustics?

Resonance occurs when a new sound note of frequency matches with standard audio frequency. Sometimes the window - panel, sections of the wooden portion are thrown into vibrations to produce new sounds, which results in interference between original sound and created sound. This will create disturbance to the audience**.**

9. State the applications of liquid penetrate test.

a. The system is used in the aerospace industries for the quality control of production and regular maintenance and safety checks.
b. Typical components which are checked by this system are turbine rotor discs and blades, aircraft wheels, castings components and welded assemblies.

10. What is the principle used for finding the velocity of ultrasonics using acoustical grating?

When ultrasonics are passed through a liquid like Kerosene contained in a tank, due to variation in pressure, the liquid act as acoustical grating, it produces different orders of spectrum due to diffraction. Using diffraction condition we can find the velocity of ultrasonics $v = \nu\lambda$ Where ν = frequency of ultrasonics λ = Wavelength of ultrasonic

11. Are the ultrasonic waves are electromagnetic waves? Give proper reasons to your answer.

Ultrasonic waves are mechanical waves. They do not possess electric and magnetic vectors as in electromagnetic waves. They need medium for propagation and also they are longitudinal in nature. They travel with the velocity of sound. Hence they are not electromagnetic

12. Define Doppler effect.

There is an apparent change in frequency of the sound waves emitted from the source when there is a relative motion between the source and observer. This effect is called Doppler Effect and the shift in frequency is called Doppler shift.

13. How can we control reverberation time?

a. By providing windows and ventilators which can be opened and closed to make the value of reverberation time optimum
b. Using heavy curtains with folds
c. Lining the walls with absorbent materials such as felt, glass, wool etc.,
d. By covering the floor with carpet
e. By providing acoustics tiles.

14. How will you ensure adequate loudness in a hall?

a. Using large sounding boards behind the speaker
b. By providing low ceiling for the reflection of energy towards the audience
c. By providing additional energy with the help of equipments like loudspeakers

15. How will you ensure uniform distribution of sound energy in the hall?

a. By taking care that there are no curved surfaces. If such ones are present they should be covered with sound absorbent materials
b. By having low ceiling

16. What is meant by Non-Destructive testing (NDT)?

Without damaging the component, if it is tested than the testing method is called as non-destructive testing (NDT) method.

17. Name some important NDT methods.

Important NDT methods are

a. Liquid penetrant method
b. Ultrasonic flaw detector
c. X-ray radiography
d. Thermography

18. What are the advantages of a non-destructive testing of products?

a. They provide the information on the quality of the material or component without damaging.
b. Presence of defects can be detected both at the manufacturing stage as well as during service
c. This helps to prevent failures in service and results in not only financial saving but saving human lives also

19. What is liquid penetrant method?

Liquid penetrant testing is one of the simplest non-destructive methods of finding discontinuities or openings in the surface of a non –porous solid material.

20. What is the principle used in liquid penetrant method?

A liquid (penetrant) is applied to the cleaned surface of the specimen and sufficient time allowed for penetration of the liquid into thedefect by capillary action. After sufficient time has passed for the penetrant to enter the discontinuity, the surface of the part is cleaned. Capillary action is again employed to act as a blotter to draw penetrant from the discontinuity by applying developer.

21. What are the essential steps in liquid penetrant inspection?

There are five essential steps in the liquid penetrant inspection methods they are

a. Surface preparation
b. Application of penetrant
c. Removal of excess penetrant
d. Development
e. Observation and Inspection.

PROBLEMS

1. Calculate the reverberation time of a hall of volume 15000m^3. The average absorption coefficient of absorbing surfaces is 0.3 sabine/m^2. The total absorbing surface area is 24000 m^2. (May 2017)

Given:

Volume of hall, V = 15000 m^3

Absorption coefficient, a = 0.3 sabine/m^2

surface area, s = 24000 m^2

Reverberation time, T = $\frac{0.167V}{as} = \frac{0.167\times15000}{0.3\times24000}$=0.3479 sec

2. A hall has a volume of 12500 m^3 and reverberation time of 1.5 sec. If 200 cushion chairs are additionally placed in the hall, what will be the new reverberation time of the hall? The absorption of each chair is 1 O.W.U.

Given:

Volume of hall, V = 12500 m^3

addition of cushioned chairs $T_1 = \frac{0.167V}{\sum as}$

Total absorption ($\sum as$) $= \frac{0.167V}{T} = \frac{0.167\times12500}{1.5}$ = 1391 OWU.m^2 or sabine

Reverberation time with addition of 200 cushioned chairs

$T_2 = \frac{0.167V}{\sum as + a_2 s_2} = \frac{0.167\times12500}{1391+200} = 1.32$ sec i.e., $a_2 s_2$ = 200 OWU.m^2

3. Calculate the frequency of the fundamental note emitted by a piezoelectric crystal of vibrating length 3mm. E= 8×10^{10} Nm^{-2} and density 2500 kgm^{-3}

Given:

Vibrating length of the crystal, t = 3mm = 0.003m

Young's modulus, E = 8×10^{10} Nm^{-2}

Density of the crystal, ρ = 2500 kgm^{-3}

For fundamental mode, P = 1

The frequency of the length vibrations, f = $\frac{P}{2t}\sqrt{\frac{E}{\rho}}$

$$f_1 = \frac{P}{2t}\sqrt{\frac{E}{\rho}} = \frac{1}{2\times0.003}\sqrt{\frac{8\times10^{10}}{2500}} = 32\times 10^6 \text{ Hz}$$

4. Calculate the frequency to which a piezo electric oscillator circuit shoud be tuned so that a Piezo –electric crystal of thickness 0.1cm vibrates in its fundamental mode to generate ultrasonic waves.(Young's modulus and density of material of crystal are 80 Gpa and 2654Kgm^{-3}

Given:

Thickness, t = 0.1×10^{-2}m

For fundamental mode, P = 1

Young's modulus, E = 80G pa

Density of the material, ρ = 2654Kgm^{-3}

The frequency of vibration, f = $\frac{p}{2t}\sqrt{\frac{E}{\rho}} = \frac{1}{2\times10^{-2}}\sqrt{\frac{80\times10^{-9}}{2654}}$ = 2.7451×10^{6} Hz

5. A quartz crystal of thickness 0.001m vibrates in its fundamental frequency. Calculate its frequency .Given that E= 7.9 $\times$ 10^{10}N/m^2 and ρ = 2650 Kgm^{-3} for quartz(jan2011)

Given:

Thickness, t = 1×10^{-3}m

For fundamental mode, P = 1

Young's modulus, E = $7.9\times 10^{10} N/m^2$

Density of the material, ρ = 2650Kgm^{-3}

The frequency of vibration is given by f = $\frac{p}{2t}\sqrt{\frac{E}{\rho}} = \frac{1}{2\times10^{-3}}\sqrt{\frac{7.9\times10^{10}}{2650}}$

= 2.7299×10^{6} Hz

6. Find the depth of a submerged submarine if an ultrasonic wave is received after 0.33 sec. from the time of transmission .The velocity of ultrasonic waves in sea water is 1440m/s

Given:

Velocity of Ultrasonic wave, v =1440m/s

Time taken between transmitted and received ultrasonic waves, t = 0.33 sec

Distance travelled, D =v × t = 1440× 0.33 = 475.2 m

Therefore depth of the submerged submarine = 475.2/2 = 237.3m

7. Longitudinal standing waves are set up in a quartz plate with antinodes at opposite faces. The fundamental frequency of vibration is given by the relation f = 2.87 × 10^3 /t where f is in Hz and t is in metre. Compute (i) Young's modulus of quartz plate and (ii) The thickness of the plate required for a frequency of 1300 KHz. The density of quartz is 2660 kg m^{-3}

Given:

Fundamental mode, P	=	1
Frequency of vibration, f	=	2.87×10^3 /t
Density of quartz, ρ	=	2660 kg m^{-3}

(i) The frequency of vibration is given by $f = \frac{P}{2t}\sqrt{\frac{E}{\rho}}$

$$\frac{2.87\times10^3}{t} = \frac{1}{2t}\sqrt{\frac{E}{2660}}$$

$$E = 4 \times 2660 \times (2.87\times10^3)^2 = 8.76406 \times 10^{10}\ Nm^{-2}$$

(ii). The frequency of vibration =1300 KHz

$$f = \frac{P}{2t}\sqrt{\frac{E}{\rho}}, \text{ Hence the Thickness (t)} = \frac{1}{2f}\sqrt{\frac{E}{\rho}}$$

$$= \frac{1}{2\times1300\times10^{3}}\sqrt{\frac{8.76406\times10^{10}}{2660}} = 2.2076 \times 10^{-3}m$$

Pondicherry University (Part – B Questions)

1. **Give a detailed account on the basic principle, the construction and production of ultrasonic waves using Magnetostriction oscillator. (Jan10, Jan11 Jan13, Jan14, Jan16 April 11, May12, 17)**

   Answer: Refer page No:3 to Page No: 5

2. **Write an essay about the acoustics of buildings and the factors affecting it. (Jan13, Jan14, April 10, May11, May12)**

   Answer: Refer page No:11 to Page No: 15

3. **Explain with neat sketch the construction and production of ultrasonic waves using Piezoelectric oscillator. (Jan14, 16 May11,13,17)**

   Answer: Refer page No:5 to Page No: 8

4. **Derive Sabine's mathematical relation for reverberation time (Jan10, Jan11, Jan15, May13)**

   Refer page No:17 to Page No: 24

5. **Calculate the frequency of the fundamental note and the first overtone emitted by a piezoelectric crystal, using the following data. Vibrating length = 5mm; young's modulus = 7.9×10^{10} Nm^{-2} and density of crystal = 2650 kgm^{-3}? (Jan11)**

Answer: Refer page No:40

6. **Give the liquid penetrant method of surface analysis (Jan14, 15, 16)**

Answer: Refer page No:26 to Page No: 29

7. **Give the pulse echo method of ultrasonic waves (Jan13, 14, 16)**

Answer: Refer page No:24 to Page No: 26

8. **With theory, explain the method of determination of velocity of ultrasonic waves in liquid using acoustic grating (Jan15)**

Answer: Refer page No:8 to Page No:9

9. **Explain Doppler Effect in light and its applications to radar.**

Answer: Refer page No:29 to Page No: 31